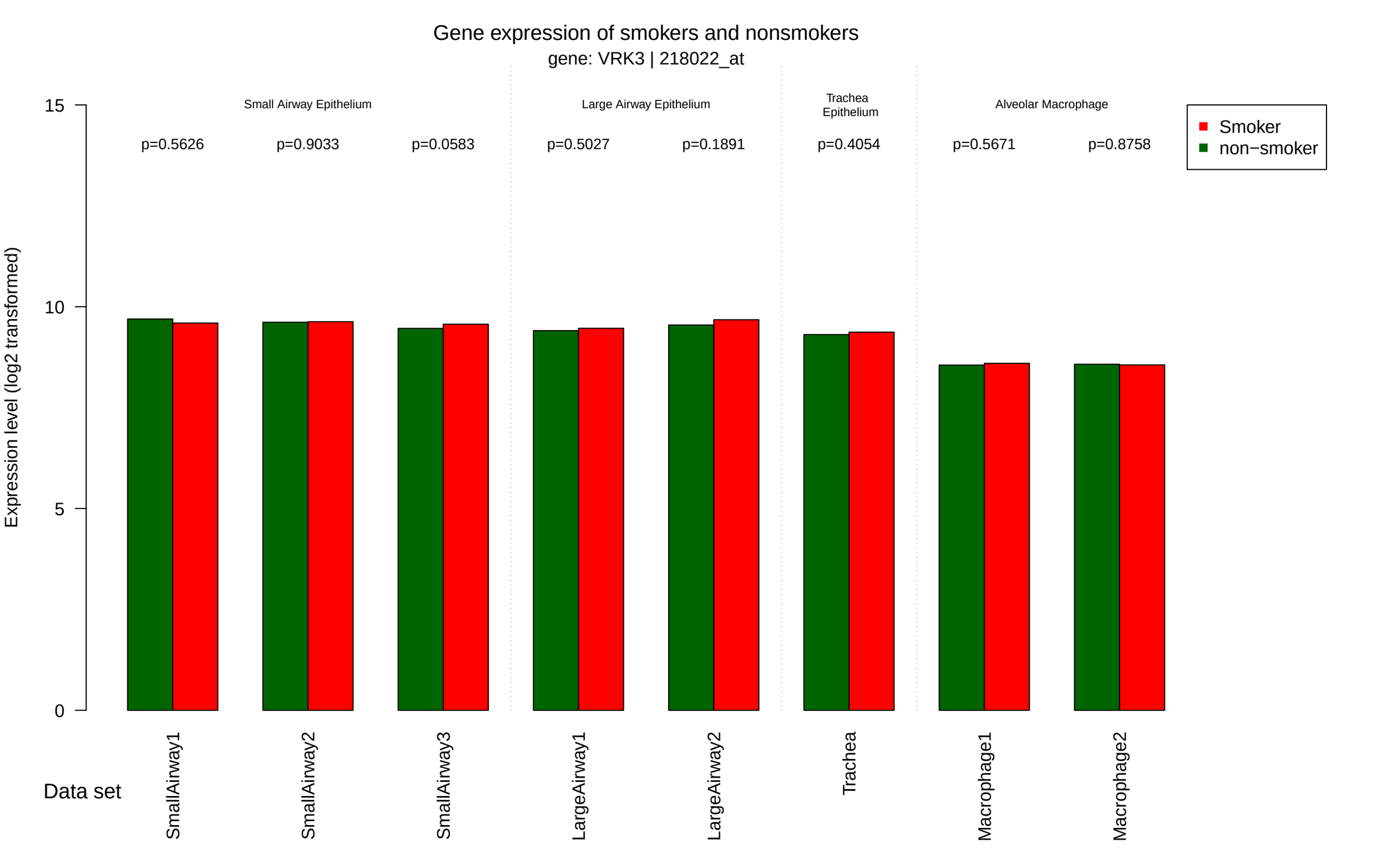
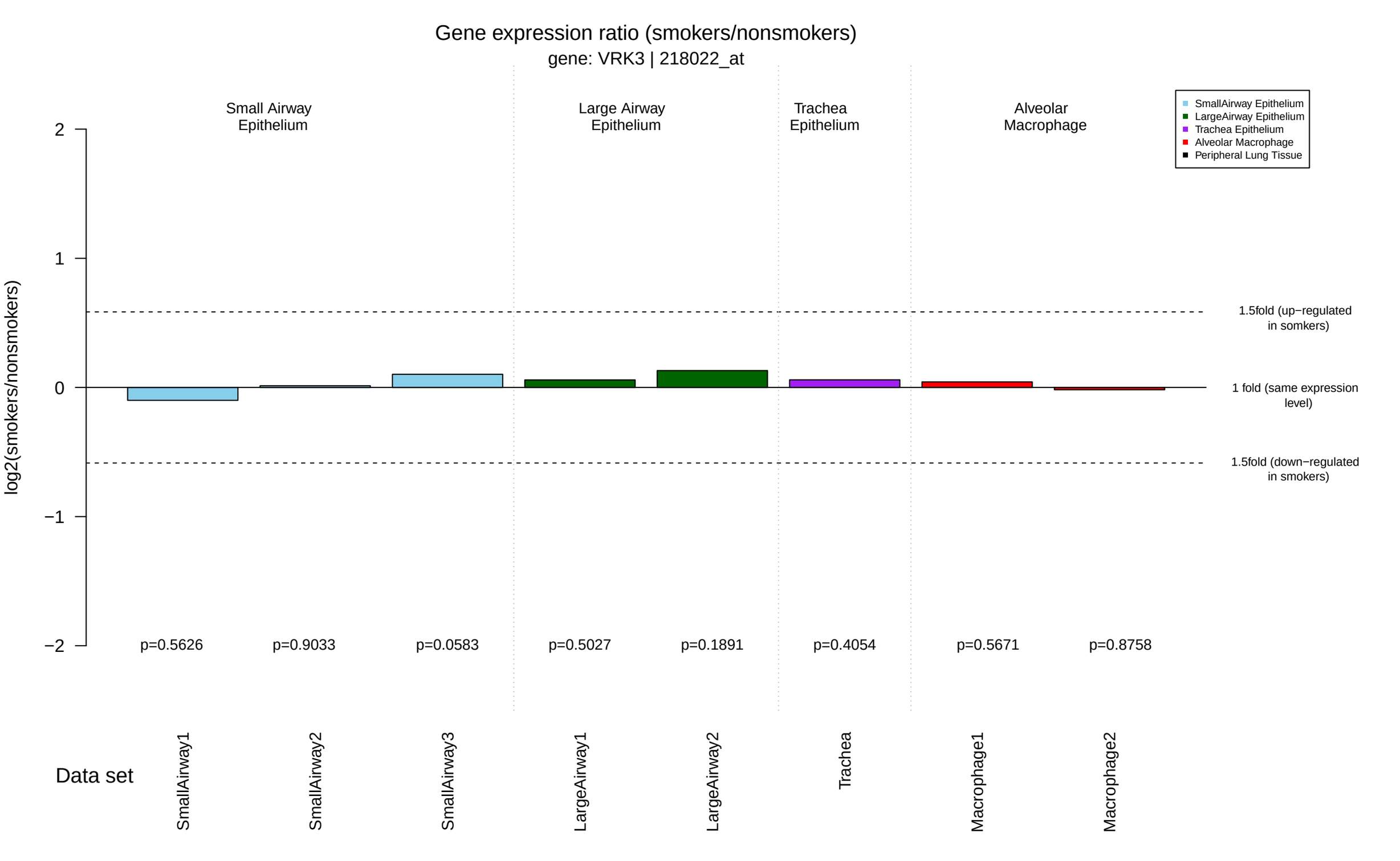
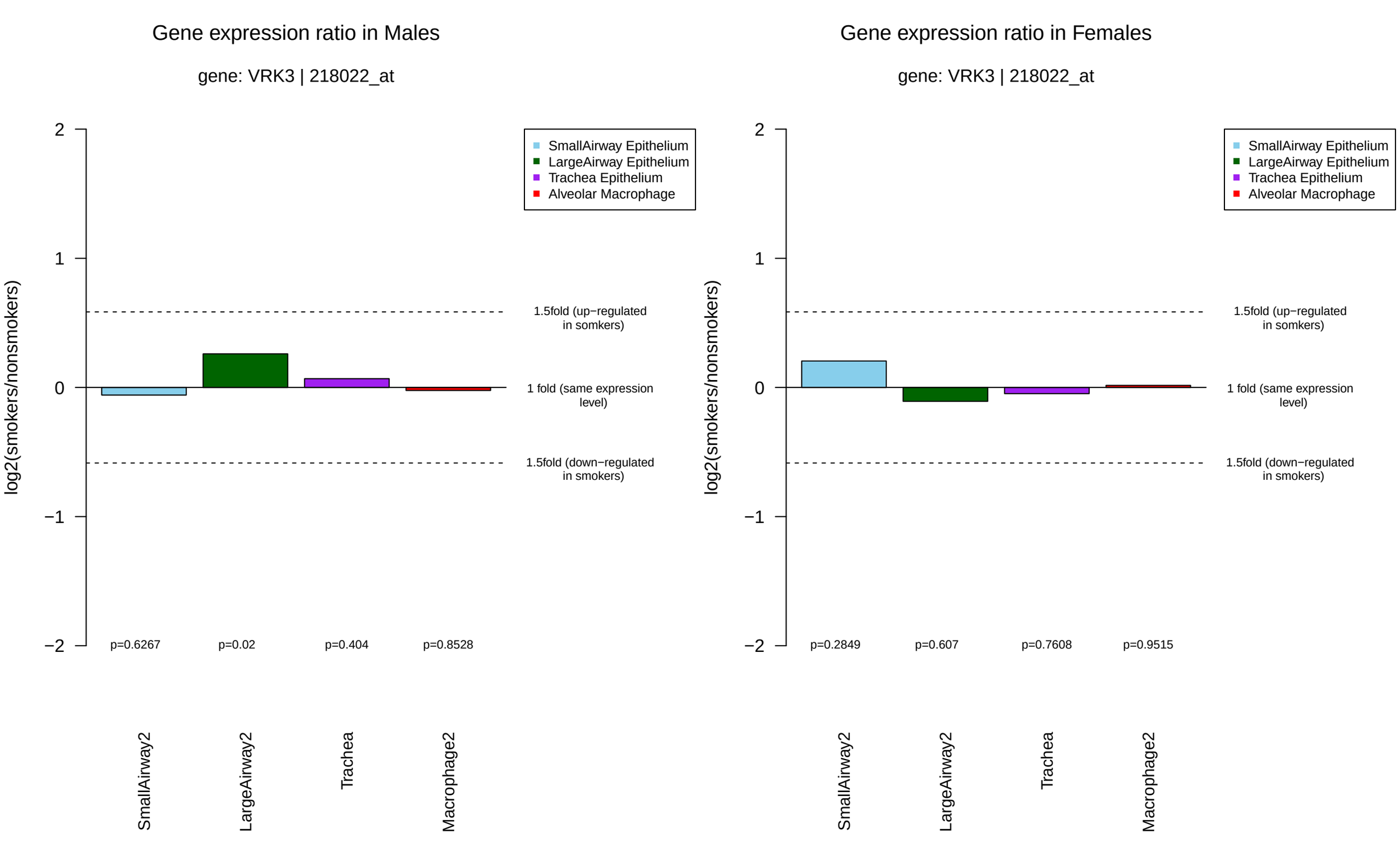
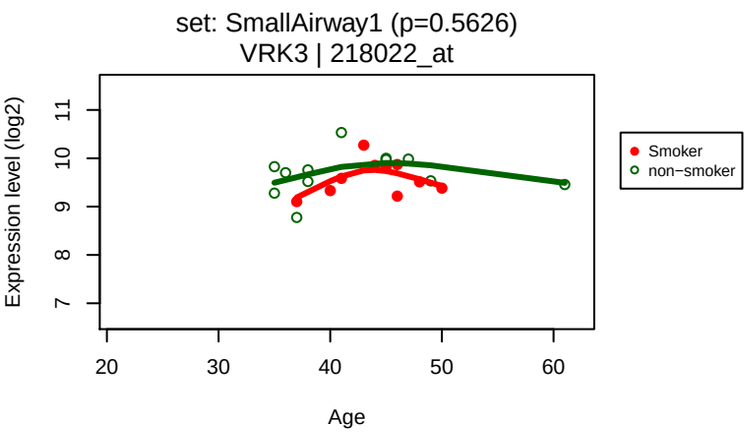
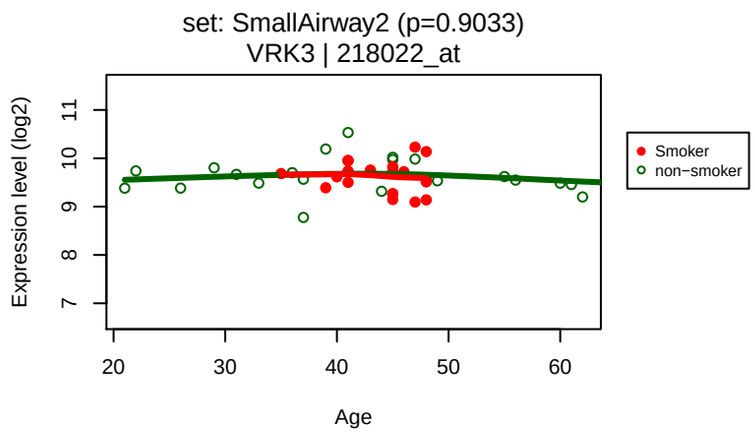
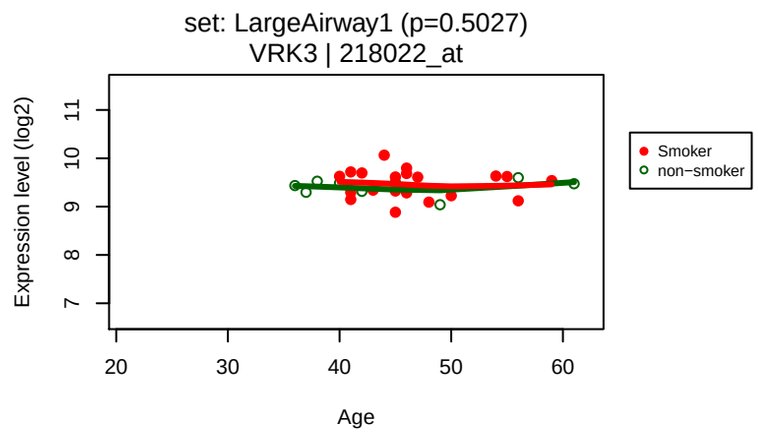

Gene expression of smokers and nonsmokers
gene: VRK3 | 218022_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5626
p=0.9033
p=0.0583
p=0.5027
p=0.1891
p=0.4054
p=0.5671
p=0.8758
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

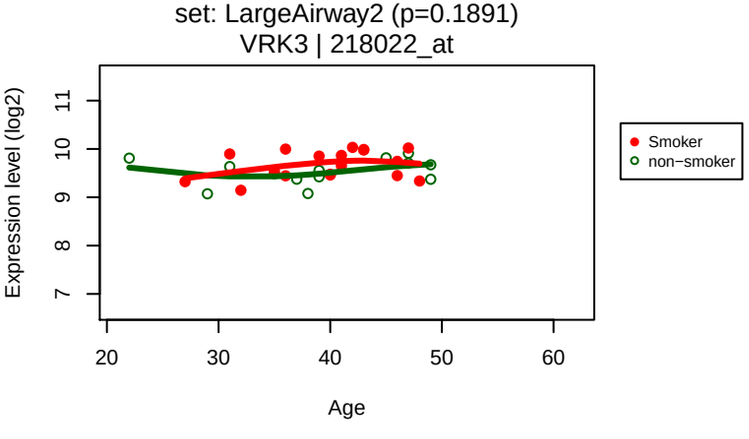

Gene expression ratio (smokers/nonsmokers)
gene: VRK3 | 218022_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5626
p=0.9033
p=0.0583
p=0.5027
p=0.1891
p=0.4054
p=0.5671
p=0.8758
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

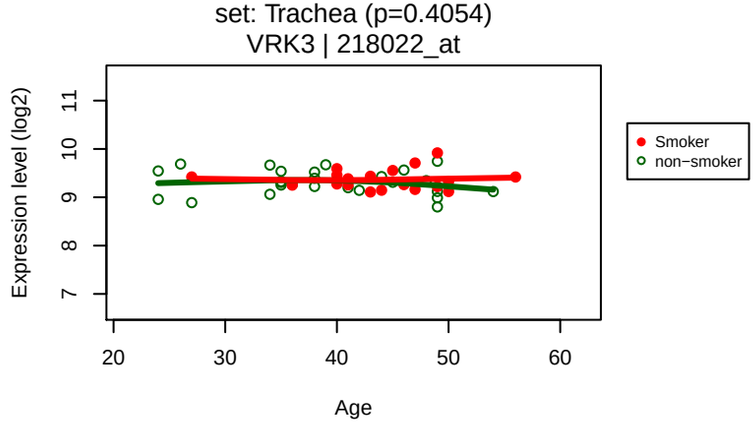

Gene expression ratio in Males
gene: VRK3 | 218022_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6267
p=0.02
p=0.404
p=0.8528
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VRK3 | 218022_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2849
p=0.607
p=0.7608
p=0.9515
SmallAirway2
LargeAirway2
Trachea
Macrophage2

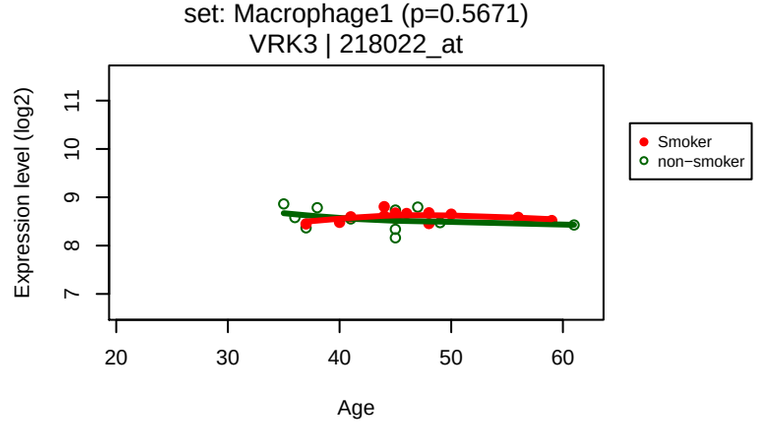

set: SmallAirway1 (p=0.5626)
VRK3 | 218022_at
Expression level (log2)
Age
Smoker
non−smoker

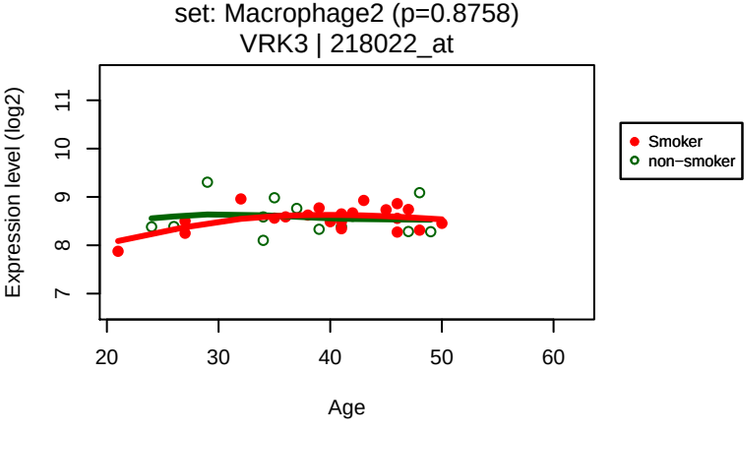

set: SmallAirway2 (p=0.9033)
VRK3 | 218022_at
Expression level (log2)
Age
Smoker
non−smoker

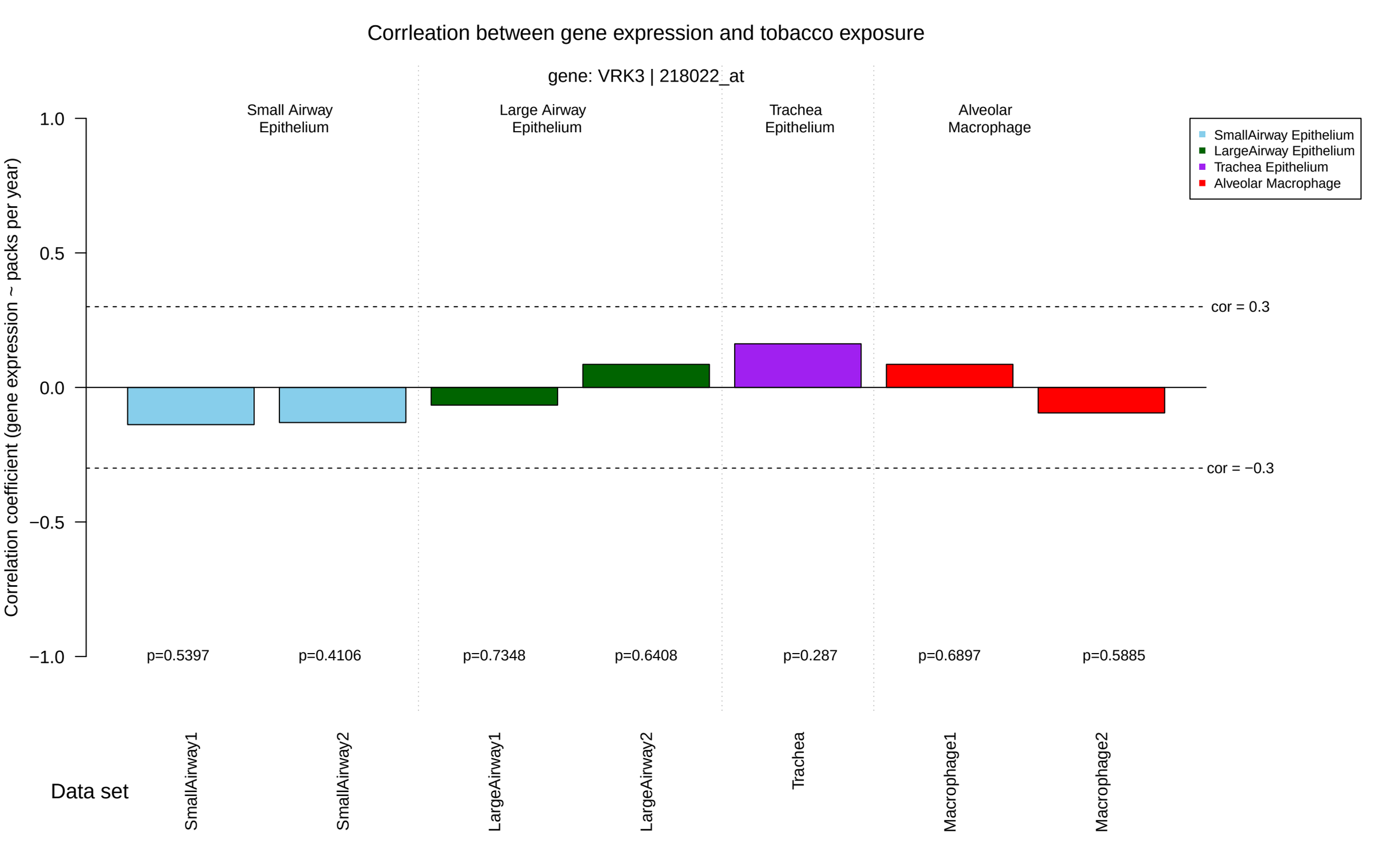

set: LargeAirway1 (p=0.5027)
VRK3 | 218022_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1891)
VRK3 | 218022_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4054)
VRK3 | 218022_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5671)
VRK3 | 218022_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8758)
VRK3 | 218022_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VRK3 | 218022_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5397
p=0.4106
p=0.7348
p=0.6408
p=0.287
p=0.6897
p=0.5885
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2